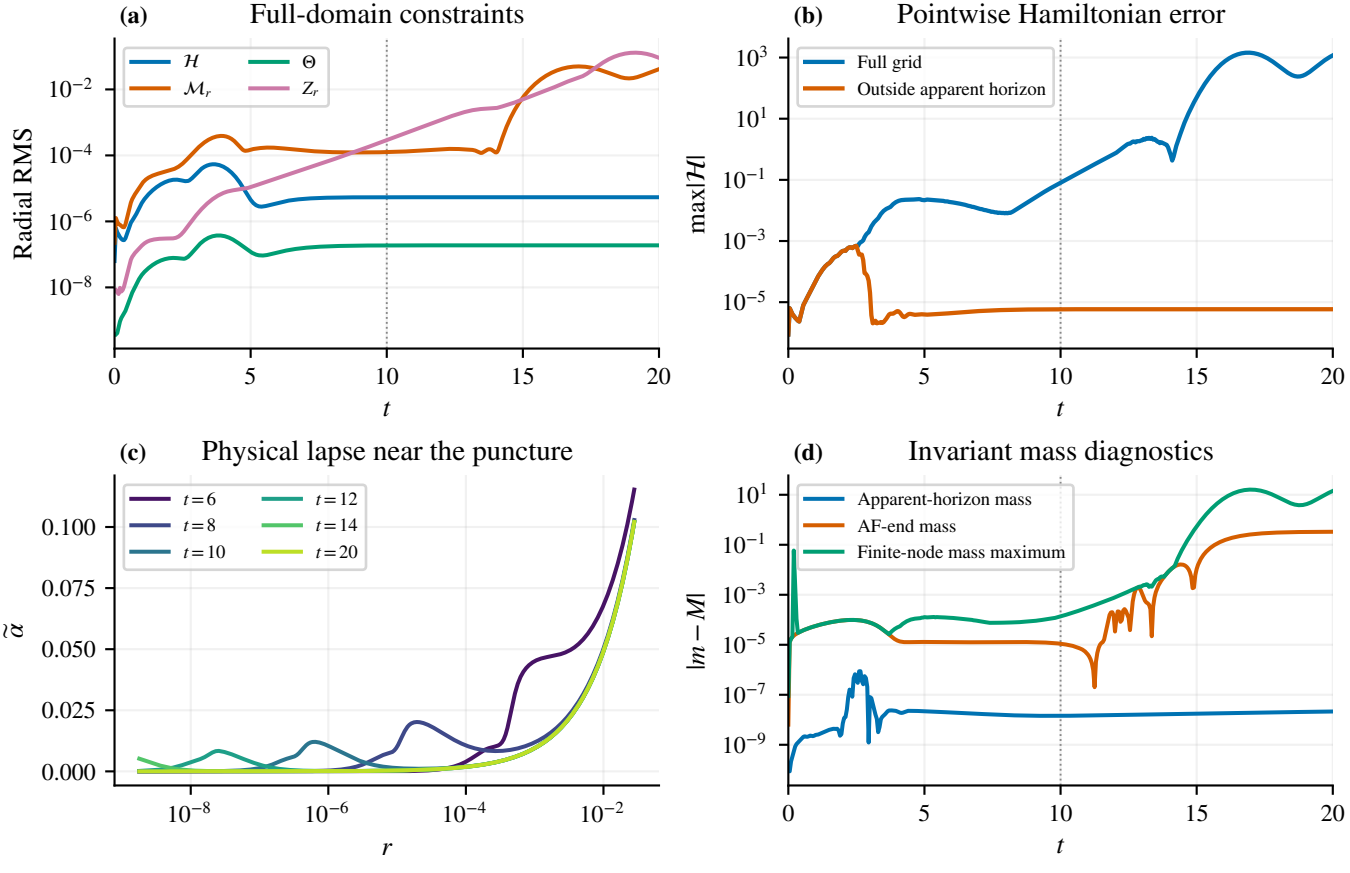
$N = 1025$ puncture pilot: target $t = 500$, evidence through $t = 20$
(a)
Full-domain constraints
$\mathcal{H}$
$\mathcal{M}_r$
Θ
Z_r
Radial RMS
10^{-2}
10^{-4}
10^{-6}
10^{-8}
0
5
10
15
20
t
(b)
Pointwise Hamiltonian error
Full grid
Outside apparent horizon
$\max|\mathcal{H}|$
10^{3}
10^{1}
10^{-1}
10^{-3}
10^{-5}
0
5
10
15
20
t
(c)
Physical lapse near the puncture
$t = 6$
$t = 8$
$t = 10$
$t = 12$
$t = 14$
$t = 20$
$\tilde{\alpha}$
0.100
0.075
0.050
0.025
0.000
10^{-8}
10^{-6}
10^{-4}
10^{-2}
r
(d)
Invariant mass diagnostics
Apparent-horizon mass
AF-end mass
Finite-node mass maximum
$|m - M|$
10^{1}
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
0
5
10
15
20
t